

Ehsan Sharafian

Ph.D. Candidate, Department of Mechanical Engineering, University of Maine, U.S.

✉ ehsan.sharafian@maine.edu ☎ (207) 631-6979 🌐 ehsansharafian.com in LinkedIn 🎓 Google Scholar 🐙 GitHub

PROFESSIONAL SUMMARY

Mechanical Engineering Ph.D. candidate seeking a Summer 2027 internship in wearable technology. Experienced in developing integrated wearable systems for gait analysis and rehabilitation using IMU sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback. Collaborative researcher and graduate student leader with experience coordinating multidisciplinary projects and human-subject studies involving 85+ participants.

EDUCATION

Ph.D. in Mechanical Engineering University of Maine, U.S. GPA: 4.00/4.00 Dissertation: Smart Real-Time Gait Training System	Sep 2023 - Dec 2027
M.S. in Mechanical Engineering Amirkabir University of Technology (Tehran Polytechnic) GPA: 3.65/4.00 Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces	Sep 2017 - Feb 2020
B.S. in Mechanical Engineering University of Tehran GPA: 3.43/4.00 Thesis: Design and Optimization of an Ankle Exoskeleton	Sep 2012 - Jul 2017

TECHNICAL SKILLS

Programming & Data Analysis	Python, Java, C/C++, NumPy, Scikit-learn, PyTorch, SPSS
Mobile & Software	MATLAB, Simulink, TwinCAT, Android Studio, Unity, Git
Wearable & Embedded Systems	IMU sensors, Arduino IDE, SparkFun modules, Sensor fusion, PCB modules
Biomechanics & Signal Processing	Gait analysis, Human activity recognition, Feature extraction, OpenSim
Robotics	Surgical robotics, Parallel manipulators, Inverse kinematics, Dynamics, Control systems
CAD	SolidWorks

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant Biorobotics and Biomechanics Lab, University of Maine, U.S.	Sep 2023 - Present
- Designed a smartphone-deployed system using a foot-mounted IMU to reconstruct foot trajectories and classify walking conditions in real time, achieving 96% classification accuracy, <1.1 cm trajectory error, and 112 ms latency for mobile gait analysis. - Developed a lightweight dual-IMU activity-recognition classifier for monitoring mobility in older adults, achieving 99% accuracy for normal locomotion and 93% accuracy for atypical stair-negotiation patterns. - Architected a home-based haptic gait-training system integrating IMUs, vibrotactile motors, custom Android applications, and PCB circuitry for adults aged 65+, improving walking speed by up to 30% with minimal cognitive demand. - Conducted human-subject validation studies involving 85+ participants, coordinating data collection and analysis to evaluate wearable and mobile systems for gait analysis and rehabilitation.	
Mechanical and Control Systems Engineer Sina Robotics and Medical Innovators Co.	Dec 2022 - Sep 2023
- Developed PLC-based control logic in Beckhoff TwinCAT and validated inverse kinematics in Simulink for a telesurgical laparoscopic robotic platform, supporting precise and reliable manipulator control. - Built a Python graphical user interface on a Raspberry Pi for surgeon-facing robot control during testing, providing real-time system guidance and warnings to improve human-robot interaction. - Coordinated hands-on validation sessions with 6 experienced surgeons, translating usability feedback and troubleshooting findings into refined surgeon-facing robotic workflows.	

ACADEMIC EXPERIENCE

Teaching Assistant | University of Maine, U.S.

Sep 2023 - Dec 2023

- Instructed and graded SolidWorks projects for a Manufacturing Design course, reinforcing practical manufacturing concepts for undergraduate students.

Teaching Assistant | Amirkabir University of Technology

Nov 2020 - Nov 2022

- Supported instruction in Dynamics II and Dynamics of Machines across **5 semesters**, helping students build foundational knowledge of mechanical systems.
- Mentored **6 undergraduate students** across five research teams, guiding equipment use, technical problem-solving, and project execution.
- Designed and assembled robotics and mechatronics setups integrating hardware, CAD, and MATLAB, enabling hands-on experimentation for undergraduate projects.

Mathematics Lecturer | Allame Tabatabayi High School

Sep 2012 - Dec 2017

- Taught mathematics to **40+ students across five classes**, strengthening analytical reasoning and problem-solving skills foundational to STEM education.

HONORS AND AWARDS

Flagship Fellowship Award, University of Maine

Sep 2024 - Sep 2025

Awarded in recognition of outstanding academic performance and research achievement.

Graduate Student Government Award for conference proposal

Mar 2025

Awarded in the **top 10%** of 100 applicants for a conference proposal on a human activity recognition system for the American Society of Biomechanics (ASB) 2025 Conference.

LEADERSHIP & SERVICE

Graduate Student Senator, Department of Mechanical Engineering, University of Maine

Sep 2025 - Present

- Advocate for graduate student concerns in senate meetings, strengthening communication between students, faculty, and university leadership.

President, Persian Graduate Student Association, University of Maine

Sep 2024 - Apr 2026

- Organized Persian cultural events, including Nowruz celebrations with **100+ attendees**, to introduce and promote Persian traditions within the university community.
- Directed event logistics and team collaboration, coordinating preparations for cultural and community programs.

PUBLICATIONS

- Ehsan Sharafian**, Babak Hejrati. "Real-Time Foot Height Estimation and Activity Classification Using a Foot-Mounted IMU Implemented on a Smartphone." *Sensors* (2026).
- Ehsan Sharafian**, Colby Ellis, Babak Hejrati. "Real-Time Activity Recognition Using Minimal Biomechanical Features: A Lightweight IMU-Based Classifier for Older Adults." *IEEE Access* (2025).
- Ehsan Sharafian**, Colby Ellis, Ben Sidaway, Marie Hayes, Babak Hejrati. "The Effects of Real-Time Haptic Feedback on Gait and Cognitive Load in Older Adults." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2025).
- Mohsen Noghani, **Ehsan Sharafian**, Ben Sidaway, Babak Hejrati. "Increasing Thigh Extension with Haptic Feedback Affects Leg Coordination in Young and Older Adult Walkers." *Journal of Biomechanics* (2025).
- Ehsan Sharafian**, Maryam Ghassabzadeh, Afshin Taghvaeipour. "Trajectory Optimization of a Spot-Welding Robot in the Joint and Cartesian Spaces." *International Journal of Robotics and Automation* (2023).
- Ehsan Sharafian**, Afshin Taghvaeipour, Maryam Ghassabzadeh. "Revisiting Screw Theory-Based Approaches in the Constraint Wrench Analysis of Robotic Systems." *Robotica* (2022).